

CHAPTER # 01: CELL AND SUB CELLULAR ORGANELLES

(Notes Chapter 1)

MCQs:

1. Which structure is primarily responsible for maintaining the selective permeability of a cell?

- A. Cell wall
- B. Plasma membrane
- C. Cytoplasm
- D. Nuclear envelope

Answer: B. Plasma membrane

2. A cell contains 70S ribosomes, circular DNA, and no membrane-bound nucleus. It is most likely:

- A. Animal cell
- B. Plant cell
- C. Prokaryotic cell
- D. Fungal cell

Answer: C. Prokaryotic cell

3. The fluid mosaic model describes the plasma membrane as:

- A. A rigid protein sheet
- B. A phospholipid bilayer containing proteins
- C. A cellulose membrane
- D. A carbohydrate bilayer

Answer: B. A phospholipid bilayer containing proteins

4. Which organelle receives proteins from the rough ER and modifies and sorts them?

- A. Lysosome
- B. Golgi complex
- C. Peroxisome
- D. Ribosome

Answer: B. Golgi complex

5. The main structural component of the plant cell wall is:

- A. Glycogen
- B. Chitin
- C. Cellulose
- D. Peptidoglycan

Answer: C. Cellulose

6. Which organelle contains hydrolytic enzymes capable of digesting macromolecules?

- A. Lysosome
- B. Ribosome
- C. Centriole

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D. Chloroplast

Answer: A. Lysosome

7. The folded structures of the inner mitochondrial membrane are called:

A. Grana

B. Cisternae

C. Cristae

D. Thylakoids

Answer: C. Cristae

8. A signal molecule that cannot cross the plasma membrane generally interacts with:

A. A cytoplasmic ribosome

B. A membrane receptor

C. Nuclear DNA

D. Mitochondrial DNA

Answer: B. A membrane receptor

9. Which organelle is particularly involved in the detoxification of hydrogen peroxide?

A. Golgi complex

B. Peroxisome

C. Ribosome

D. Nucleolus

Answer: B. Peroxisome

10. The region of a prokaryotic cell containing its chromosome is called the:

A. Nucleolus

B. Nucleoid

C. Nucleus

D. Centrosome

Answer: B. Nucleoid

11. Which structure is directly responsible for protein synthesis?

A. Ribosome

B. Golgi complex

C. Lysosome

D. Vacuole

Answer: A. Ribosome

12. The cis face of the Golgi apparatus generally:

A. Releases secretory vesicles

B. Receives vesicles from the ER

C. Produces ATP

D. Synthesizes DNA

Answer: B. Receives vesicles from the ER

13. The Calvin cycle takes place primarily in the:

A. Thylakoid membrane

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- B. Chloroplast stroma
- C. Mitochondrial matrix
- D. Cytoplasm

Answer: B. Chloroplast stroma

14. Which organelle is particularly abundant in cells involved in lipid synthesis?

- A. Smooth ER
- B. Rough ER
- C. Lysosome
- D. Nucleolus

Answer: A. Smooth ER

15. Which process releases substances from a cell through fusion of vesicles with the plasma membrane?

- A. Endocytosis
- B. Exocytosis
- C. Phagocytosis
- D. Pinocytosis

Answer: B. Exocytosis

16. The membrane surrounding the central vacuole of a plant cell is called:

- A. Plasmalemma
- B. Tonoplast
- C. Nuclear envelope
- D. Crista

Answer: B. Tonoplast

17. Which organelle contains chlorophyll and is specialized for photosynthesis?

- A. Chromoplast
- B. Chloroplast
- C. Amyloplast
- D. Mitochondrion

Answer: B. Chloroplast

18. The first stage of a typical cell-signaling pathway is:

- A. Response
- B. Reception
- C. Translation
- D. Replication

Answer: B. Reception

19. Which molecule can act as a second messenger in cell signaling?

- A. cAMP
- B. Cellulose
- C. DNA
- D. Glycogen

Answer: A. cAMP

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20. The nucleolus is mainly involved in:

- A. ATP synthesis
- B. Ribosomal subunit formation
- C. Lipid digestion
- D. Photosynthesis

Answer: B. Ribosomal subunit formation

21. Which organelle has its own DNA and ribosomes?

- A. Lysosome
- B. Golgi apparatus
- C. Mitochondrion
- D. Vacuole

Answer: C. Mitochondrion

22. Glyoxisomes are particularly important in:

- A. Germinating oil-rich seeds
- B. Mature animal erythrocytes
- C. Neurons
- D. Bacteria

Answer: A. Germinating oil-rich seeds

23. A protein destined for secretion from a eukaryotic cell is generally synthesized on:

- A. Free ribosomes only
- B. Ribosomes attached to rough ER
- C. Lysosomes
- D. Centrioles

Answer: B. Ribosomes attached to rough ER

24. Which process involves uptake of extracellular material by the plasma membrane?

- A. Exocytosis
- B. Endocytosis
- C. Translation
- D. Transcription

Answer: B. Endocytosis

25. The microtubules of a typical centriole are arranged as:

- A. Nine triplets
- B. Nine doublets
- C. Nine singlets
- D. Ten triplets

Answer: A. Nine triplets

26. Which structure is absent as a membrane-bound organelle in prokaryotes?

- A. Ribosome
- B. Plasma membrane
- C. Nucleus

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D. Cytoplasm

Answer: C. Nucleus

27. The aqueous internal medium in which many cellular reactions occur is the:

A. Cytosol

B. Cell wall

C. Nucleoid

D. Thylakoid

Answer: A. Cytosol

28. Which organelle is responsible for intracellular digestion?

A. Lysosome

B. Golgi complex

C. Ribosome

D. Chloroplast

Answer: A. Lysosome

29. Which type of plastid is primarily associated with storage of starch?

A. Chromoplast

B. Amyloplast

C. Chloroplast

D. Elaioplast

Answer: B. Amyloplast

30. The light-dependent reactions of photosynthesis occur mainly on the:

A. Chloroplast envelope

B. Thylakoid membranes

C. Stroma

D. Outer mitochondrial membrane

Answer: B. Thylakoid membranes

31. Which organelle is structurally composed of flattened membrane-bound sacs called cisternae?

A. Golgi complex

B. Ribosome

C. Centriole

D. Nucleolus

Answer: A. Golgi complex

32. Which protein removes phosphate groups from phosphorylated proteins?

A. Protein kinase

B. Protein phosphatase

C. Catalase

D. Ligase

Answer: B. Protein phosphatase

33. Protein kinases generally function by:

- A. Removing phosphate groups
- B. Adding phosphate groups
- C. Breaking DNA
- D. Digesting proteins

Answer: B. Adding phosphate groups

34. Which organelle is bounded by a double membrane and contains thylakoids?

- A. Lysosome
- B. Chloroplast
- C. Peroxisome
- D. Golgi complex

Answer: B. Chloroplast

35. A cell that lacks mitochondria, ER, Golgi apparatus, and a nucleus is most likely:

- A. Prokaryotic
- B. Plant
- C. Animal
- D. Fungal

Answer: A. Prokaryotic

36. The plasma membrane is mainly composed of:

- A. Phospholipids and proteins
- B. Cellulose and starch
- C. DNA and RNA
- D. Glycogen and cellulose

Answer: A. Phospholipids and proteins

37. Receptor-mediated endocytosis is characterized by:

- A. Non-specific uptake only
- B. Specific ligand-receptor interaction
- C. Direct movement through DNA
- D. Cell wall degradation

Answer: B. Specific ligand-receptor interaction

38. Which organelle is associated with beta-oxidation of very-long-chain fatty acids?

- A. Peroxisome
- B. Ribosome
- C. Nucleolus
- D. Golgi apparatus

Answer: A. Peroxisome

39. Catalase converts hydrogen peroxide primarily into:

- A. Glucose and ATP
- B. Water and oxygen
- C. Carbon dioxide and glucose

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D. Pyruvate and oxygen

Answer: B. Water and oxygen

40. Which organelle contains enzymes for lipid metabolism and detoxification but lacks ribosomes?

A. Smooth ER

B. Rough ER

C. Ribosome

D. Nucleolus

Answer: A. Smooth ER

41. A granum is a stack of:

A. Cristae

B. Thylakoids

C. Cisternae

D. Ribosomes

Answer: B. Thylakoids

42. The fluid surrounding the grana in a chloroplast is the:

A. Matrix

B. Stroma

C. Cytosol

D. Nucleoplasm

Answer: B. Stroma

43. Which structure regulates communication between the nucleus and cytoplasm?

A. Nuclear pore complex

B. Cristae

C. Tonoplast

D. Cell wall

Answer: A. Nuclear pore complex

44. Chromatin is primarily composed of:

A. DNA and associated proteins

B. Lipids and carbohydrates

C. RNA and cellulose

D. ATP and proteins

Answer: A. DNA and associated proteins

45. Which cell type normally has a cell wall made of peptidoglycan?

A. Plant cell

B. Bacterial cell

C. Animal cell

D. Human red blood cell

Answer: B. Bacterial cell

46. Which structure provides mechanical support to plant cells outside the plasma membrane?

- A. Cell wall
- B. Ribosome
- C. Golgi apparatus
- D. Cytosol

Answer: A. Cell wall

47. Which organelle is most directly associated with the synthesis of ribosomal RNA?

- A. Nucleolus
- B. Lysosome
- C. Peroxisome
- D. Golgi apparatus

Answer: A. Nucleolus

48. The main function of transport vesicles is to:

- A. Carry cargo between cellular compartments
- B. Replicate DNA
- C. Produce ribosomes
- D. Generate chromosomes

Answer: A. Carry cargo between cellular compartments

49. Which process involves engulfment of relatively large particles?

- A. Pinocytosis
- B. Phagocytosis
- C. Exocytosis
- D. Osmosis

Answer: B. Phagocytosis

50. Pinocytosis is commonly described as cellular:

- A. Eating
- B. Drinking
- C. Respiration
- D. Digestion

Answer: B. Drinking

51. Which statement correctly distinguishes eukaryotic cells from prokaryotic cells?

- A. Eukaryotes lack DNA
- B. Eukaryotes contain membrane-bound organelles
- C. Prokaryotes lack ribosomes
- D. Prokaryotes have membrane-bound nuclei

Answer: B. Eukaryotes contain membrane-bound organelles

52. Which structure is found in both prokaryotic and eukaryotic cells?

- A. Nucleus
- B. Golgi complex
- C. Ribosome

D. Mitochondrion

Answer: C. Ribosome

53. A signal molecule binds a receptor and activates intracellular proteins. This process is called:

A. Signal transduction

B. Translation

C. Replication

D. Glycolysis

Answer: A. Signal transduction

54. Steroid hormones commonly exert their effects by binding receptors:

A. Inside the target cell

B. On the cell wall

C. Inside lysosomes

D. On ribosomes

Answer: A. Inside the target cell

55. Which sequence best represents the basic organization of cell signaling?

A. Response → reception → transduction

B. Reception → transduction → response

C. Transduction → replication → reception

D. Translation → reception → digestion

Answer: B. Reception → transduction → response

56. Which feature of the plasma membrane is mainly responsible for its fluidity?

A. Cellulose fibers

B. Lipid composition and molecular movement

C. DNA molecules

D. Peptidoglycan

Answer: B. Lipid composition and molecular movement

57. Which organelle is the major site of ATP production through aerobic respiration?

A. Mitochondrion

B. Golgi apparatus

C. Lysosome

D. Vacuole

Answer: A. Mitochondrion

58. The mitochondrial matrix contains enzymes primarily involved in:

A. Citric acid cycle

B. Light reactions

C. Protein secretion

D. Cell wall synthesis

Answer: A. Citric acid cycle

59. Which structure increases the surface area of the inner mitochondrial membrane?

- A. Grana
- B. Cristae
- C. Cisternae
- D. Microvilli

Answer: B. Cristae

60. Which plastid gives many flowers and fruits their characteristic yellow, orange, or red coloration?

- A. Chromoplast
- B. Amyloplast
- C. Chloroplast
- D. Proplastid

Answer: A. Chromoplast

61. Stem cells are characterized by their ability to:

- A. Divide and differentiate into specialized cell types
- B. Perform photosynthesis only
- C. Lack DNA permanently
- D. Function only as red blood cells

Answer: A. Divide and differentiate into specialized cell types

62. A stem cell capable of producing many different specialized cell types within a particular developmental range is called:

- A. Pluripotent
- B. Acellular
- C. Non-dividing
- D. Terminally differentiated

Answer: A. Pluripotent

63. Which structure is directly involved in maintaining plant cell turgidity?

- A. Central vacuole
- B. Ribosome
- C. Nucleolus
- D. Centriole

Answer: A. Central vacuole

64. Which organelle contains digestive enzymes but is itself protected by a membrane?

- A. Lysosome
- B. Ribosome
- C. Nucleolus
- D. Centriole

Answer: A. Lysosome

65. Which structure connects the rough ER to the nucleus?

- A. Nuclear envelope
- B. Cell wall

C. Tonoplast

D. Crista

Answer: A. Nuclear envelope

66. Ribosomes are composed primarily of:

A. rRNA and proteins

B. DNA and lipids

C. Cellulose and RNA

D. Lipids and carbohydrates

Answer: A. rRNA and proteins

67. The rough ER is directly associated with:

A. Protein synthesis and processing

B. Photosynthetic carbon fixation

C. DNA replication

D. ATP storage

Answer: A. Protein synthesis and processing

68. Which organelle is part of the endomembrane system?

A. Golgi apparatus

B. Mitochondrion

C. Chloroplast

D. Centriole

Answer: A. Golgi apparatus

69. A secretory vesicle releases its contents when it:

A. Fuses with the plasma membrane

B. Fuses with DNA

C. Enters the nucleus

D. Breaks down the cell wall

Answer: A. Fuses with the plasma membrane

70. Which organelle can contain oxidative enzymes and hydrogen peroxide?

A. Peroxisome

B. Ribosome

C. Golgi complex

D. Nucleolus

Answer: A. Peroxisome

71. Glyoxisomes are considered specialized forms of:

A. Peroxisomes

B. Ribosomes

C. Lysosomes

D. Mitochondria

Answer: A. Peroxisomes

72. The glyoxylate cycle is particularly useful because it permits conversion of:

- A. Acetyl-CoA into carbohydrate precursors
- B. DNA into ATP
- C. Proteins directly into cellulose
- D. ATP into DNA

Answer: A. Acetyl-CoA into carbohydrate precursors

73. Which organelle has an outer and inner membrane with an intermembrane space?

- A. Mitochondrion
- B. Lysosome
- C. Golgi apparatus
- D. Peroxisome

Answer: A. Mitochondrion

74. The inner membrane of chloroplasts contains the:

- A. Thylakoid system
- B. Nuclear pore system
- C. Golgi cisternae
- D. Lysosomal system

Answer: A. Thylakoid system

75. ATP and NADPH produced during the light-dependent reactions are primarily used in:

- A. Calvin cycle
- B. Glycolysis
- C. DNA replication
- D. Lysosomal digestion

Answer: A. Calvin cycle

76. Which structure is mainly responsible for organizing microtubules during animal-cell division?

- A. Centrosome
- B. Lysosome
- C. Golgi apparatus
- D. Vacuole

Answer: A. Centrosome

77. Centrioles are particularly associated with:

- A. Microtubule organization
- B. Protein digestion
- C. Photosynthesis
- D. Lipid storage

Answer: A. Microtubule organization

78. Which organelle is absent from mature mammalian red blood cells?

- A. Nucleus
- B. Plasma membrane
- C. Cytoplasm

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D. Proteins

Answer: A. Nucleus

79. The nuclear envelope contains openings called:

- A. Nuclear pores
- B. Cristae
- C. Grana
- D. Plasmodesmata

Answer: A. Nuclear pores

80. The primary function of nuclear pores is to:

- A. Regulate transport between nucleus and cytoplasm
- B. Produce ATP
- C. Digest proteins
- D. Synthesize cellulose

Answer: A. Regulate transport between nucleus and cytoplasm

81. Which process can amplify an extracellular signal inside a cell?

- A. Phosphorylation cascade
- B. Cell wall synthesis
- C. DNA packaging
- D. Ribosome degradation

Answer: A. Phosphorylation cascade

82. Which molecule commonly acts as an extracellular signaling molecule?

- A. Hormone
- B. Ribosome
- C. Histone
- D. Cellulose

Answer: A. Hormone

83. If a membrane receptor becomes permanently inactive, the cell would most directly lose its ability to:

- A. Receive the corresponding external signal
- B. Produce DNA
- C. Form ribosomes
- D. Maintain its cytoplasm

Answer: A. Receive the corresponding external signal

84. Which type of signaling usually acts on cells relatively close to the signaling cell?

- A. Paracrine signaling
- B. Endocrine signaling
- C. Long-distance neuronal signaling only
- D. Intracellular signaling

Answer: A. Paracrine signaling

85. Endocrine signaling commonly involves signals transported through:

- A. Blood or body fluids
- B. Ribosomes
- C. Cell walls
- D. Nuclear pores only

Answer: A. Blood or body fluids

86. Which organelle is responsible for sorting proteins into different destinations after modification?

- A. Golgi complex
- B. Mitochondrion
- C. Centriole
- D. Ribosome

Answer: A. Golgi complex

87. A lysosomal enzyme is synthesized initially on:

- A. Rough ER-bound ribosomes
- B. Smooth ER only
- C. Free ribosomes only
- D. Centrioles

Answer: A. Rough ER-bound ribosomes

88. The Golgi apparatus modifies proteins mainly by:

- A. Enzymatic processing and addition/removal of molecular groups
- B. Replicating their DNA
- C. Producing ribosomes
- D. Converting them directly into ATP

Answer: A. Enzymatic processing and addition/removal of molecular groups

89. Which organelle is especially important for intracellular calcium storage in some cells?

- A. Smooth ER
- B. Lysosome
- C. Golgi apparatus
- D. Nucleolus

Answer: A. Smooth ER

90. Which cellular process requires vesicle formation from the plasma membrane?

- A. Endocytosis
- B. Exocytosis
- C. Translation
- D. Transcription

Answer: A. Endocytosis

91. During exocytosis, the vesicle membrane becomes:

- A. Incorporated into the plasma membrane
- B. Incorporated into nuclear DNA
- C. Converted into a ribosome

D. Destroyed by the nucleolus

Answer: A. Incorporated into the plasma membrane

92. A major difference between a cell wall and plasma membrane is that the cell wall:

- A. Is generally rigid and provides structural support
- B. Controls all molecular transport selectively
- C. Contains phospholipid bilayer as its main component
- D. Is found in all animal cells

Answer: A. Is generally rigid and provides structural support

93. Which structure provides additional protection and shape to many prokaryotic cells?

- A. Cell wall
- B. Nucleolus
- C. Golgi apparatus
- D. Mitochondrial matrix

Answer: A. Cell wall

94. Which of the following is a non-membranous structure?

- A. Ribosome
- B. Lysosome
- C. Peroxisome
- D. Golgi complex

Answer: A. Ribosome

95. Which organelle contains enzymes for cellular detoxification and lipid metabolism?

- A. Peroxisome
- B. Ribosome
- C. Nucleolus
- D. Centriole

Answer: A. Peroxisome

96. Which structure is most directly responsible for the selective movement of ions across the plasma membrane?

- A. Membrane transport proteins
- B. Cellulose fibers
- C. Nuclear pores
- D. Ribosomal RNA

Answer: A. Membrane transport proteins

97. If a cell has chloroplasts, a large central vacuole, and a cellulose cell wall, it is most likely a:

- A. Plant cell
- B. Bacterial cell
- C. Animal cell
- D. Viral particle

Answer: A. Plant cell

98. Which feature strongly supports the endosymbiotic origin of mitochondria and chloroplasts?

- A. Their own DNA and ribosomes
- B. Their lack of membranes
- C. Their presence in all prokaryotes
- D. Their production of lysosomal enzymes

Answer: A. Their own DNA and ribosomes

99. Which statement about eukaryotic cells is correct?

- A. Their DNA is always free in the cytoplasm
- B. They possess a membrane-bound nucleus
- C. They lack ribosomes
- D. They lack membrane-bound organelles

Answer: B. They possess a membrane-bound nucleus

100. If a secretory cell is treated so that its Golgi apparatus cannot function, which process would be most directly affected?

- A. Protein modification and sorting
- B. DNA replication
- C. Glycolysis
- D. Chromosome formation

Answer: A. Protein modification and sorting

1. Why is the plasma membrane described as selectively permeable rather than completely permeable?

Answer:

The plasma membrane permits some substances to cross more easily than others. Small non-polar molecules can often pass through the lipid bilayer relatively easily. Ions and many polar molecules require specific transport proteins. Therefore, the membrane controls the movement of substances according to cellular requirements.

2. How does the structure of the phospholipid bilayer contribute to membrane function?

Answer:

Phospholipids possess hydrophilic heads and hydrophobic tails. The heads face the aqueous environments while the tails form the membrane's interior. This hydrophobic core restricts the passage of many ions and polar molecules. Membrane proteins provide selective pathways for substances that cannot cross directly.

3. Why are ribosomes attached to rough ER important for secretion of proteins?

Answer:

Ribosomes attached to rough ER synthesize proteins entering the ER lumen or membrane. These proteins may be destined for secretion, lysosomes, or cellular membranes. The ER provides an environment for folding and initial protein modification. The proteins can subsequently be transported to the Golgi through vesicles.

4. Why can a defect in the Golgi apparatus affect several different cellular functions?

Answer:

The Golgi modifies, sorts, and packages many proteins and lipids. It directs these molecules toward different cellular destinations. Defective Golgi activity can therefore affect secretion, lysosomal enzyme targeting, and membrane composition. Thus, the Golgi functions as a major distribution center of the endomembrane system.

5. How does the mitochondrion's inner membrane facilitate ATP production?

Answer:

The inner membrane contains the electron transport chain and ATP synthase. Electron transfer generates a proton gradient across this membrane. Protons flow back through ATP synthase, providing energy for ATP formation. Cristae increase the membrane surface available for these processes.

6. Why are chloroplast thylakoid membranes essential for photosynthesis?

Answer:

Thylakoid membranes contain chlorophyll and other photosynthetic components. They contain the electron transport chains involved in light-dependent reactions. Light energy drives electron transfer and contributes to formation of a proton gradient. The resulting ATP and NADPH are used in carbon fixation reactions.

7. Explain why prokaryotic cells can perform essential cellular functions without membrane-bound organelles.

Answer:

Prokaryotic cells possess a plasma membrane, cytoplasm, ribosomes, and genetic material. Their metabolic processes occur in the cytoplasm or at the plasma membrane. They do not require a membrane-bound nucleus for genetic information storage.

Their simpler organization allows essential processes to occur without membrane-bound organelles.

8. Why is the nucleoid not considered a true nucleus?

Answer:

The nucleoid contains the major chromosome of a prokaryotic cell. However, it is not surrounded by a nuclear envelope. A true nucleus is enclosed by a double membrane containing nuclear pores. Therefore, the nucleoid lacks the membrane-bound organization characteristic of eukaryotic nuclei.

9. Why are lysosomes sometimes called the digestive compartments of cells?

Answer:

Lysosomes contain numerous hydrolytic enzymes. These enzymes break down proteins, lipids, carbohydrates, and nucleic acids. They can digest material taken into the cell and damaged cellular components. Their acidic interior provides suitable conditions for many lysosomal enzymes.

10. Why does the destruction of lysosomal membranes potentially damage a cell?

Answer:

Lysosomes contain powerful hydrolytic enzymes. Normally, these enzymes are separated from the cytoplasm by the lysosomal membrane. If membrane integrity is severely disrupted, enzymes may enter the cytoplasm. This can contribute to degradation of cellular components.

11. Why are peroxisomes important in protecting cells from oxidative damage?

Answer:

Many peroxisomal reactions produce hydrogen peroxide. Hydrogen peroxide can damage cellular molecules if it accumulates. Peroxisomes contain catalase and other enzymes involved in peroxide metabolism. Catalase converts hydrogen peroxide into water and oxygen.

12. How are glyoxisomes especially useful to germinating oil-rich seeds?

Answer:

Oil-rich seeds contain large quantities of stored lipids. Glyoxisomes contain enzymes involved in the glyoxylate cycle. This pathway allows carbon from fatty-acid breakdown to contribute to carbohydrate production.

The resulting carbohydrates can support growth before extensive photosynthesis begins.

13. Why does a plant cell become turgid when water enters its central vacuole?

Answer:

Water enters the vacuole through osmosis when appropriate water-potential conditions exist.

The expanding vacuole pushes the cytoplasm against the cell wall. The rigid cell wall resists excessive expansion.

The resulting internal pressure contributes to cell rigidity and plant support.

14. How does the central vacuole contribute to plant-cell homeostasis?

Answer:

The vacuole stores water, ions, pigments, metabolites, and waste materials. Its membrane regulates movement of substances between the cytoplasm and vacuole. Water storage contributes to maintenance of turgor pressure. Thus, the vacuole participates in both storage and osmotic regulation.

15. Why are mitochondria and chloroplasts considered semiautonomous organelles?

Answer:

Both contain their own DNA and ribosomes. They can synthesize some of their own proteins. However, many proteins required by these organelles are encoded by nuclear genes. Therefore, they are genetically independent to some extent but still depend on the nucleus.

16. Why does the chloroplast have two different major internal environments, the stroma and thylakoid system?

Answer:

The thylakoid membranes contain the machinery for light-dependent reactions. The stroma contains enzymes involved mainly in carbon fixation.

Separating these environments allows different reactions to occur under appropriate conditions.

This compartmentalization increases the efficiency of photosynthesis.

17. How does the Golgi apparatus maintain functional organization despite continuous vesicle traffic?

Answer:

The Golgi has distinct cis and trans regions. Vesicles from the ER generally enter through the cis side. Cargo is modified as it moves through successive Golgi compartments. The trans-Golgi network sorts cargo into vesicles for specific destinations.

18. Why are transport vesicles important for maintaining compartmentalization in eukaryotic cells?

Answer:

Different organelles maintain different chemical environments. Transport vesicles move selected cargo between these compartments. The vesicles can fuse with specific target membranes. This allows substances to be transported without mixing the entire contents of organelles.

19. How does receptor-mediated endocytosis provide specificity to cellular uptake?

Answer:

Specific extracellular molecules bind to complementary receptors on the plasma membrane. Receptor binding concentrates selected molecules in membrane regions. These regions then form vesicles and enter the cell. Therefore, uptake can occur selectively rather than through completely random engulfment.

20. Why is exocytosis important for cells that secrete large molecules?

Answer:

Large molecules generally cannot cross the phospholipid bilayer directly. They can instead be packaged into membrane-bound secretory vesicles. The vesicle fuses with the plasma membrane during exocytosis. Its contents are released outside while the vesicle membrane becomes part of the plasma membrane.

21. Why can a cell respond to one signaling molecule but another nearby cell may not respond?

Answer:

A target cell must possess the appropriate receptor for a signal. Cells lacking that receptor cannot normally detect the signal in the same way. Receptor activation initiates intracellular signal-transduction pathways. Therefore, receptor distribution contributes strongly to cellular specificity.

22. How can a small extracellular signal produce a large intracellular response?

Answer:

One activated receptor can activate several intracellular signaling molecules. These molecules can activate additional proteins in a cascade. Second messengers can further amplify the signal. Consequently, a relatively small external signal can produce a substantial cellular response.

23. Why are protein kinases and phosphatases both important in cell signaling?

Answer:

Protein kinases add phosphate groups to target proteins. Phosphorylation can alter the activity, location, or interactions of proteins. Protein phosphatases remove phosphate groups and can reverse these effects. Together they provide regulated control over signaling pathways.

24. How does cAMP function as a second messenger?

Answer:

A signal received at the cell surface can activate an enzyme that produces cAMP. cAMP then carries the signal within the cell. It can activate protein kinase A and initiate phosphorylation of target proteins. This allows amplification and coordination of the cellular response.

25. Why can steroid hormones often use intracellular receptors?

Answer:

Steroid hormones are lipid-soluble molecules.

They can pass through the hydrophobic interior of the plasma membrane. Inside the cell, they bind receptors located in the cytoplasm or nucleus. The resulting hormone-receptor complex can influence gene expression.

26. Why is the nucleus considered the control center of many eukaryotic cells?

Answer:

The nucleus contains most of the cell's genetic information. Genes provide instructions for production of many cellular proteins. Regulation of gene expression therefore influences cellular structure and activity. The nuclear envelope also regulates exchange between the nucleus and cytoplasm.

27. What would be the likely consequence if nuclear pores stopped functioning?

Answer:

Transport between the nucleus and cytoplasm would become severely disrupted. RNA and certain proteins would not move normally between these compartments. Gene expression and ribosome-related processes could therefore be affected. Ultimately, many cellular functions would be impaired.

28. Why is the nucleolus essential even though it does not have a surrounding membrane?

Answer:

The nucleolus is a specialized region within the nucleus. It is the major site of rRNA synthesis and ribosomal subunit assembly. Ribosomes are essential for protein synthesis throughout the cell. Thus, the nucleolus plays a critical role despite being non-membranous.

29. How do centrioles contribute to cell division?

Answer:

Centrioles are associated with centrosomes that organize microtubules. During cell division, microtubules form the spindle apparatus. The spindle helps organize and separate chromosomes. Therefore, centriole-associated microtubule organization contributes to accurate cell division in many animal cells.

30. Why are ribosomes described as molecular machines?

Answer:

Ribosomes coordinate the translation of genetic information from mRNA. They position mRNA and tRNAs accurately during protein synthesis. They catalyze formation of peptide bonds between amino acids. Thus, they convert nucleotide-based information into a polypeptide sequence.

31. Why can smooth ER abundance vary between different cell types?

Answer:

The amount of smooth ER reflects the metabolic requirements of a cell. Cells specialized in lipid synthesis require extensive smooth ER. Cells involved in detoxification may also contain abundant smooth ER. Therefore, organelle abundance is related to cellular specialization.

32. Why do highly active cells generally contain numerous mitochondria?

Answer:

Active cells require large quantities of ATP. Mitochondria produce ATP through aerobic respiration. Greater mitochondrial abundance can provide greater capacity for ATP production. Examples include cells with high energy demands such as muscle cells.

33. Why does increasing the surface area of an organelle's membrane often increase its functional capacity?

Answer:

Many membrane-dependent reactions require proteins embedded in membranes. A larger membrane surface provides more space for these proteins. This can increase the number of reactions occurring simultaneously. Cristae in mitochondria and thylakoids in chloroplasts demonstrate this principle.

34. Why are prokaryotic ribosomes generally described as 70S while cytoplasmic eukaryotic ribosomes are 80S?

Answer:

The S value represents sedimentation behavior rather than simple addition of molecular masses.

Prokaryotic ribosomes consist of 50S and 30S subunits. Cytoplasmic eukaryotic ribosomes consist of 60S and 40S subunits. Their different structures and compositions produce different sedimentation coefficients.

35. Why can antibiotics that target bacterial ribosomes affect bacteria more than human cytoplasmic ribosomes?

Answer:

Bacterial ribosomes differ structurally from eukaryotic cytoplasmic ribosomes. Many antibiotics recognize features specific to bacterial ribosomes. This can selectively interfere with bacterial protein synthesis. The difference provides a basis for selective antimicrobial action.

36. Why is compartmentalization important in eukaryotic cells?

Answer:

Different organelles provide specialized environments for different reactions. This prevents incompatible reactions from interfering with one another. It also allows enzymes and substrates to be concentrated efficiently. Therefore, compartmentalization increases regulation and metabolic efficiency.

37. How does the endomembrane system coordinate protein processing and transport?

Answer:

Ribosomes on rough ER synthesize proteins entering the secretory pathway. The ER folds and initially modifies these proteins. Transport vesicles deliver them to the Golgi complex. The Golgi modifies, sorts, and packages them for their final destinations.

38. Why does endocytosis require changes in the plasma membrane?

Answer:

The cell must enclose extracellular material within a membrane-bound compartment. The plasma membrane therefore bends inward around the material. The membrane eventually pinches off to form an intracellular vesicle. This allows extracellular substances to enter without directly passing through the lipid bilayer.

39. Differentiate phagocytosis and pinocytosis on a conceptual basis.

Answer:

Phagocytosis involves uptake of relatively large solid particles or cells. Pinocytosis involves uptake of extracellular fluid and dissolved substances. Both are forms of endocytosis and involve plasma-membrane invagination. Their major difference is the type and scale of material internalized.

40. Why can endocytosis and exocytosis alter the surface area of the plasma membrane?

Answer:

During endocytosis, portions of the plasma membrane are internalized into vesicles. This tends to reduce the surface area of the plasma membrane. During exocytosis, vesicle membranes fuse with the plasma membrane. This adds membrane material to the cell surface.

41. Why are stem cells important in multicellular organisms?

Answer:

Stem cells can undergo self-renewal and produce differentiated cell types. They contribute to development and tissue maintenance. Their differentiation potential allows production of specialized cells. Therefore, stem cells are important for growth, repair, and biological research.

42. Distinguish pluripotency from multipotency.

Answer:

Pluripotent cells can differentiate into many cell types derived from the major embryonic lineages.

Multipotent cells have a more restricted differentiation potential. They generally produce several related cell types within a particular tissue or lineage. Thus, pluripotency represents broader developmental potential than multipotency.

43. Why is differentiation not simply an increase in cell size?

Answer:

Differentiation involves changes in gene expression and cellular specialization. Different genes become active or inactive in different cell types.

This produces specific proteins and cellular structures. Consequently, cells acquire specialized functions rather than merely becoming larger.

44. Why does the cell wall not replace the plasma membrane?

Answer:

The cell wall provides rigidity and mechanical support. It is generally more porous and does not perform the same selective transport role as the plasma membrane. The plasma membrane regulates movement of substances into and out of the cytoplasm. Both structures therefore perform complementary functions.

45. How can a cell maintain different chemical conditions inside organelles?

Answer:

Organelle membranes separate internal compartments from the cytosol. Membrane proteins selectively transport ions and molecules across these membranes. Some organelles also contain pumps that establish concentration gradients. This allows each compartment to maintain conditions suitable for its specific reactions.

46. Why is the mitochondrial matrix different from the intermembrane space?

Answer:

The two compartments are separated by the inner mitochondrial membrane. The matrix contains enzymes involved in processes such as the citric acid cycle. The intermembrane space accumulates protons during electron transport. Their different compositions are essential for oxidative phosphorylation.

47. Why does a chloroplast require both thylakoid membranes and stroma?

Answer:

Thylakoid membranes contain the machinery required for light-dependent electron transfer. These reactions generate ATP and NADPH. The stroma contains enzymes that use ATP and NADPH in carbon fixation. The two compartments therefore cooperate during photosynthesis.

48. Why can a mutation in a membrane receptor change a cell's response without changing the extracellular signal?

Answer:

The extracellular signal may remain chemically unchanged. However, the receptor may fail to bind the signal or transmit its information correctly. As a result, intracellular signal transduction may not occur normally. Therefore, receptor structure is essential for converting an external signal into a cellular response.

49. Why are organelles considered specialized rather than independent structures?

Answer:

Organelles perform particular functions but depend on other cellular components. For example, mitochondria require many proteins encoded by nuclear genes. The Golgi depends on vesicles from the ER for much of its incoming cargo. Thus, organelles operate as interconnected components of an integrated cell.

50. A cell has abundant rough ER, Golgi complexes, secretory vesicles, and mitochondria. What can be inferred about its function?

Answer:

The cell is likely specialized for producing and secreting large quantities of proteins. Rough ER provides extensive protein-synthesis capacity. The Golgi modifies and sorts the synthesized proteins. Secretory vesicles transport them to the plasma membrane, while mitochondria supply ATP for these activities

LONG QUESTIONS:

1. Explain the organization of the plasma membrane and relate its structure to its major cellular functions.

1. Phospholipid Bilayer

The plasma membrane consists mainly of a phospholipid bilayer. Hydrophilic heads face the aqueous surroundings, while hydrophobic tails form the internal region of the membrane.

2. Membrane Proteins

Integral and peripheral proteins perform transport, receptor, enzymatic, adhesion, and recognition functions. Their distribution contributes to the mosaic nature of the membrane.

3. Membrane Fluidity

Lipids and many proteins can move laterally within the membrane. This fluidity allows the membrane to change shape, form vesicles, and repair itself.

4. Selective Permeability

The hydrophobic interior restricts many ions and polar substances. Specific transport proteins provide controlled pathways for substances that cannot cross the bilayer freely.

5. Cell Signaling

Membrane receptors recognize extracellular signaling molecules. Receptor activation can initiate intracellular signal-transduction pathways and produce specific cellular responses.

6. Endocytosis and Exocytosis

The membrane can bend and form vesicles during endocytosis and fuse with vesicles during exocytosis. These processes allow bulk movement of material into and out of cells.

2. Describe the endomembrane system and explain how it coordinates synthesis, modification, and transport of proteins.

1. Rough Endoplasmic Reticulum

Ribosomes attached to rough ER synthesize proteins destined for secretion, membranes, or certain organelles. Newly synthesized proteins enter the ER where folding and initial modification occur.

2. Transport Vesicles

Proteins are packaged into transport vesicles that bud from the ER. These vesicles deliver their cargo to the Golgi complex.

3. Golgi Complex

The Golgi receives proteins at its cis face and modifies them as they pass through its cisternae. Different enzymes within the Golgi perform specific processing reactions.

4. Sorting at Trans-Golgi

The trans-Golgi region sorts processed proteins according to their destinations. Proteins may be directed toward lysosomes, the plasma membrane, or secretory vesicles.

5. Secretory Vesicles

Secretory proteins are enclosed in vesicles that move toward the plasma membrane. The vesicles protect and transport the cargo through the cytoplasm.

6. Exocytosis

The secretory vesicle fuses with the plasma membrane and releases its contents outside the cell. This completes the secretory pathway.

3. Explain the pathway of cell signaling from reception of an extracellular signal to the final cellular response.

1. Signal Molecule

A signaling molecule or ligand originates outside or within the organism and reaches a target cell. Only cells capable of detecting the signal respond specifically.

2. Reception

The ligand binds to a complementary receptor. The receptor may be located on the plasma membrane or inside the cell depending on the properties of the signaling molecule.

3. Signal Transduction

Receptor activation produces intracellular molecular changes. These may involve G proteins, protein kinases, ion channels, or other signaling components.

4. Second Messengers

Molecules such as cAMP and Ca^{2+} can relay signals within the cell. They can amplify the original signal and activate additional proteins.

5. Cellular Response

The final response may include enzyme activation, changes in gene expression, secretion, cytoskeletal changes, or cell division.

6. Signal Termination

Signaling pathways must be switched off after the appropriate response. Ligand removal, receptor inactivation, degradation of second messengers, and protein dephosphorylation can terminate signaling.

4. Compare mitochondria and chloroplasts with respect to structure, membranes, and energy transformation.

1. Basic Membrane Structure

Both mitochondria and chloroplasts are surrounded by an outer and inner membrane. Both therefore possess internal compartmentalization.

2. Mitochondrial Organization

The mitochondrial inner membrane forms cristae and encloses the matrix. Electron transport and oxidative phosphorylation occur mainly at the inner membrane.

3. Chloroplast Organization

Chloroplasts contain a stroma and an internal thylakoid membrane system. Stacks of thylakoids form grana.

4. Energy Transformation in Mitochondria

Mitochondria transfer energy from reduced molecules through electron transport chains. The resulting proton gradient drives ATP synthesis.

5. Energy Transformation in Chloroplasts

Chloroplasts capture light energy through chlorophyll-containing systems. Light-dependent reactions generate ATP and NADPH, which support carbon fixation.

6. Semiautonomous Nature

Both organelles contain DNA and ribosomes. Nevertheless, they depend extensively on proteins encoded by nuclear genes, so their autonomy is incomplete.

5. Explain the structural and functional differences between prokaryotic and eukaryotic cells.

1. Genetic Organization

Prokaryotic cells generally possess a circular chromosome located in the nucleoid. Eukaryotic cells contain chromosomes enclosed within a membrane-bound nucleus.

2. Membrane-Bound Organelles

Prokaryotes lack membrane-bound organelles such as mitochondria and Golgi complexes. Eukaryotic cells possess extensive internal membrane compartmentalization.

3. Ribosomes

Prokaryotic cytoplasmic ribosomes are generally 70S, whereas eukaryotic cytoplasmic ribosomes are generally 80S.

4. Cell Size and Complexity

Prokaryotic cells are generally smaller and structurally simpler. Eukaryotic cells are usually larger and contain specialized organelles.

5. Cell Division

Prokaryotes generally reproduce through binary fission. Eukaryotic cells use complex processes involving mitosis or meiosis depending on the biological context.

6. Functional Integration

Eukaryotic organelles divide cellular processes into specialized compartments. Prokaryotes perform their essential reactions mainly in the cytoplasm and at the plasma membrane.

6. Describe lysosomes, peroxisomes, and glyoxisomes and compare their functions.

1. Lysosomal Structure

Lysosomes are membrane-bound compartments containing hydrolytic enzymes. Their internal acidic environment supports enzyme activity.

2. Lysosomal Functions

They digest macromolecules, extracellular material taken up by endocytosis, and damaged cellular components. They therefore contribute to intracellular recycling.

3. Peroxisomal Functions

Peroxisomes contain oxidative enzymes involved in fatty-acid metabolism and detoxification. Their reactions can produce hydrogen peroxide.

4. Catalase Activity

Catalase is an important peroxisomal enzyme that converts hydrogen peroxide into water and oxygen. This helps prevent harmful accumulation of peroxide.

5. Glyoxisomes

Glyoxisomes are specialized peroxisomes particularly important in germinating oil-rich seeds. They contain enzymes involved in the glyoxylate cycle.

6. Metabolic Significance

The three organelles demonstrate different forms of cellular processing: lysosomes specialize in hydrolytic digestion, peroxisomes in oxidative metabolism, and glyoxisomes in converting stored lipid carbon toward carbohydrate production.

7. Explain how endocytosis and exocytosis maintain cellular exchange with the external environment.

1. Endocytosis

Endocytosis begins when the plasma membrane invaginates around extracellular material. The membrane then pinches off to form an intracellular vesicle.

2. Phagocytosis

Phagocytosis involves engulfment of relatively large particles or cells. It is particularly important for uptake of solid material.

3. Pinocytosis

Pinocytosis involves uptake of extracellular fluid and dissolved substances. It allows cells to internalize small amounts of surrounding fluid.

4. Receptor-Mediated Endocytosis

Specific ligands bind membrane receptors before internalization. This mechanism allows selective uptake of particular extracellular molecules.

5. Exocytosis

During exocytosis, intracellular vesicles move toward the plasma membrane and fuse with it. Their contents are released into the extracellular environment.

6. Cellular Importance

Together, endocytosis and exocytosis allow cells to exchange large quantities of material that cannot simply diffuse through the membrane. They also contribute to membrane recycling and communication.

8. Explain the relationship between organelle structure and function in a eukaryotic cell.

1. Rough ER and Protein Synthesis

Rough ER contains ribosomes that synthesize proteins entering the secretory pathway. Its membrane system provides a compartment for protein folding and processing.

2. Golgi and Protein Sorting

The Golgi receives ER-derived cargo and modifies it. Its polarized organization allows proteins to be sorted according to their destinations.

3. Mitochondria and ATP Production

Mitochondrial cristae increase the surface area of the inner membrane. This membrane contains electron-transfer components and ATP synthase.

4. Chloroplasts and Photosynthesis

Thylakoid membranes contain photosynthetic machinery, while the stroma contains enzymes for carbon fixation. Their structural separation supports efficient photosynthesis.

5. Lysosomes and Digestion

Lysosomal membranes enclose hydrolytic enzymes. The separation of these enzymes from the cytoplasm allows controlled intracellular digestion.

6. Nucleus and Genetic Control

The nucleus encloses DNA and regulates access to genetic information. Nuclear pores control exchange between nuclear and cytoplasmic compartments.

9. Explain stem cells and discuss how their properties differ from those of specialized cells.

1. Self-Renewal

Stem cells can divide while retaining a population of cells with stem-cell characteristics. This property allows long-term maintenance of stem-cell populations.

2. Differentiation

Stem cells can produce specialized cell types. Their developmental potential depends on the type of stem cell.

3. Potency

Totipotent cells have the broadest developmental potential. Pluripotent cells can produce many specialized cell types, whereas multipotent cells have a more restricted range.

4. Gene Expression

Differentiation involves changes in gene expression. Different sets of genes become active in specialized cells, producing proteins required for specific functions.

5. Tissue Maintenance

Stem cells can contribute to replacement of cells lost through normal turnover or damage in tissues where appropriate stem-cell populations exist.

6. Biological Significance

Stem cells are important for understanding development, tissue organization, regeneration, and disease mechanisms. Their properties make them valuable subjects of biological research.

10. A secretory cell contains abundant rough ER, Golgi complexes, mitochondria, and secretory vesicles. Explain how these organelles work together.

1. Protein Synthesis

Ribosomes attached to rough ER synthesize proteins destined for secretion. The polypeptides enter the ER during synthesis.

2. ER Processing

Within the ER, proteins undergo folding and initial processing. Properly processed proteins are packaged into transport vesicles.

3. Golgi Modification

Transport vesicles deliver proteins to the Golgi. The Golgi modifies the proteins and sorts them according to their final destinations.

4. Secretory Vesicles

Proteins destined for extracellular release are packaged into secretory vesicles. These vesicles transport the proteins toward the plasma membrane.

5. Exocytosis

Secretory vesicles fuse with the plasma membrane and release their contents outside the cell. Their membrane becomes incorporated into the plasma membrane.

6. ATP Supply

Mitochondria generate ATP required for many energy-demanding processes involved in protein synthesis, vesicle movement, modification, and secretion. Thus, these organelles function as an integrated system rather than independently.